

Senior Backend C#/.NET Engineer

Location: Cape Town, South Africa (Hybrid)

Company: Protium

Employment Type: Full-Time (Contract-to-Permanent Considered)

Positions Available: 2

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About the Role

Protium is seeking a highly skilled **Senior Backend C#/.NET Engineer** to design, build, and support production-grade APIs, backend services, integrations, and enterprise systems using **C#, .NET, SQL Server, and Microsoft/Azure technologies**.

This is a **Cape Town-based hybrid position**, requiring regular attendance at our Claremont office (typically 2–3 days per week). As a senior engineer, you will take ownership of key technical initiatives, contribute to architectural decisions, mentor team members, and ensure that solutions are secure, scalable, maintainable, and production-ready.

You will work on high-quality, API-driven systems where engineering excellence, security, performance, and long-term supportability are critical.

About Protium

Protium is a security-first software development company based in Claremont, Cape Town.

We specialize in building robust, scalable, and future-proof software solutions with a strong emphasis on:

- Engineering excellence
- Secure software delivery
- Practical architecture
- Clean code practices
- Long-term maintainability

Our primary focus is delivering high-quality software systems for clients in the financial services sector and other industries where reliability, security, and correctness are essential.

Working Model

- Cape Town-based
- Hybrid working arrangement
- Office collaboration 2–3 times per week
- Full-time preferred
- Contract-to-permanent considered

Please note: This is not a fully remote role. Applicants must be based in Cape Town and able to attend regular office sessions in Claremont.

Key Responsibilities

Backend Engineering & Delivery

- Design, develop, and maintain production-grade APIs and backend services using C# and .NET.
- Own backend features from planning through implementation, testing, deployment, and support.
- Build service layers, business logic, integrations, data-access components, scheduled jobs, and background services.
- Develop and optimize SQL Server databases, including schema design, indexing, performance tuning, and query optimization.

Technical Leadership

- Contribute to solution architecture and technical design discussions.
- Evaluate trade-offs and help shape implementation strategies.
- Break down complex initiatives into manageable deliverables.
- Identify risks, dependencies, performance concerns, and security implications early in the development process.
- Participate in and lead code reviews to maintain engineering standards.

Quality & Operations

- Develop and maintain automated tests for critical workflows and integrations.
- Troubleshoot and resolve issues across APIs, databases, services, and infrastructure touchpoints.
- Contribute to CI/CD pipelines, deployment processes, monitoring, observability, and operational improvements.
- Ensure systems meet high standards for security, reliability, and maintainability.

Collaboration & Mentorship

- Work closely with frontend, mobile, QA, product, design, architecture, and client-facing teams.
- Mentor junior and intermediate developers through reviews, pairing sessions, and technical guidance.
- Communicate technical decisions, risks, and progress clearly to stakeholders.

AI-Assisted Engineering

- Use AI-assisted engineering tools responsibly to accelerate delivery, improve feedback loops, and reduce repetitive work.
 - Apply strong engineering judgement to ensure correctness, maintainability, readability, testing, and security.
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Required Skills & Experience

Technical Requirements

- Strong commercial experience with **C# and .NET / ASP.NET Core**
- Proven experience building and supporting **RESTful APIs** in production environments
- Strong knowledge of **SQL Server**, relational database design, indexing, performance optimization, and data modelling
- Experience with **Entity Framework Core**, **Dapper**, or similar data-access technologies
- Solid understanding of backend architecture, layering, and separation of concerns
- Experience implementing authentication, authorization, validation, error handling, logging, and API security

Engineering Competencies

- Ability to design and implement solutions with limited supervision
- Strong understanding of software design principles, including SOLID and clean architecture concepts
- Ability to balance technical trade-offs and explain design decisions
- Strong testing, debugging, and troubleshooting capabilities
- Experience conducting code reviews and providing constructive feedback
- Ability to own features end-to-end from design through production support

Collaboration & Leadership

- Experience mentoring developers and supporting team growth
- Strong communication skills across technical and non-technical stakeholders

- Comfortable working within collaborative Git-based development workflows

Location Requirement

- Must be based in Cape Town and available for regular office collaboration in Claremont
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Core Technology Stack

Required

- C# / .NET
- ASP.NET Core
- REST APIs
- SQL Server
- Entity Framework Core and/or Dapper
- Git
- Authentication & Authorization
- Testing & Debugging
- Backend Architecture & Design Patterns

Preferred Senior-Level Experience

- Azure & Azure DevOps
 - Docker
 - CI/CD pipelines
 - Messaging and event-driven systems
 - Background/Worker Services
 - API Security
 - Logging, Monitoring & Observability
 - Performance Tuning
 - Integration-heavy Systems
 - Clean Architecture or Modular Architecture approaches
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Nice-to-Have Experience

- React, TypeScript, HTML, CSS, or other frontend technologies
- MVC, Razor Pages, Blazor, or related .NET web technologies
- Secondary backend languages such as Node.js, Java, or Python
- RabbitMQ, LavinMQ, Kafka, Azure Service Bus, or similar messaging platforms

- Event-driven architecture and pub/sub systems
 - Financial services, workflow, reporting, or secure portal platforms
 - Azure cloud hosting and infrastructure
 - Consulting, agency, or client-facing delivery environments
 - Participation in architectural and technical decision-making
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Security Mindset

While this is not a dedicated security role, candidates should demonstrate strong secure-development fundamentals and be comfortable discussing:

- Secure API development
- Authentication and authorization best practices
- Secure integration patterns
- Application security fundamentals
- Risk identification and mitigation

Security, quality, and maintainability are core principles of how Protium builds software.

Compensation & Benefits

Market-related and competitive, based on:

- Experience
- Technical depth
- Seniority
- Overall fit for the role

Additional benefits and employment structure details will be discussed during the interview process.

Why Join Protium?

- Work on meaningful, production-scale software systems
- Collaborate with experienced engineers who value quality craftsmanship
- Influence architecture and technical direction
- Mentor and help grow a strong engineering team

- Build secure, scalable software in an environment that values engineering excellence and continuous improvement